EPMA TAPP review workbook (EPMA TAPP v88) — WDS Point Analysis

This workbook asks you to review the EPMA TAPP: the list of metadata items that should be reported when an EPMA procedure is registered, and when a measurement session run under that procedure is documented. It has three tabs: Instructions (this tab), Reviewer information, and EPMA TAPP.

This copy covers WDS Point Analysis only. It shows the 82 of 95 items that apply to that mode. "#" keeps the TAPP's own numbering, so gaps in the numbers are items that belong to other modes.

Every item is described for two records. The procedure record is the standing recipe, registered once and cited by DOI. The session record describes one measurement session run under that procedure; a session may cover several samples.

What to do

Step 1	Fill in the "Reviewer information" tab. Answers go in the "Your answer" column; the questions are locked.
Step 2	On the "EPMA TAPP" tab, put all your feedback in the six shaded comment columns: Needed?, Structure right? (twice), Tier right? (twice) and Other comments. Each one judges the information column on its left; "Other comments" takes everything else. All other cells are locked and cannot be edited, so if you disagree with a name, description, example or tick, say so in the comment column beside it. Leave a cell blank if you have no view.
Step 3	Use the review criteria at the bottom of this tab.
Step 4	When you have finished, save the workbook as EPMA_TAPP_v88_Review_WDS_Point_Analysis_<your initials>.xlsx and email it to info@astromat.org.

In Group 1, the comment cells of the 14 registration fields are grey and read "Set by registration requirements; no review needed", and are locked. Those fields are common to every TAPP and are set by registration policy, so you can skip them. "Other comments" stays open on those rows, and the four coupling fields (Coupled Technique(s) to Coupled Dataset or Publication Reference) are open for review as usual.

This Instructions tab is locked too. On every tab you can still filter, resize columns and copy text.

The EPMA TAPP tab: what each column means

The tab has 8 information columns and 6 comment columns. The information columns are: "#", "Metadata item", "Description", "Example (NOT recommendations)", "Structure — procedure record", "Procedure-Level Tier", "Structure — session record", "Session-Level Tier". The comment columns are: Needed?, Structure right? (twice), Tier right? (twice) and Other comments. Each comment column sits immediately to the right of the information column it judges and has the same colour: purple for the metadata item, blue for the procedure record, yellow for the session record.

#	The item's number in the TAPP: group number, then position in the group.
Metadata item	The name of the metadata item.
Description	What the item records.
Example (NOT recommendations)	An illustration of the item filled in, from a fictional worked example (an invented EPMA session written up as a methods section). It shows the shape of an answer. It is not recommended content, and the laboratory, samples and results are invented.

Structure — procedure record	How the procedure states the item: one value, one value for each entry of a list, or a list of its own. The session record follows it, as shown in the session structure column.
Procedure-Level Tier	Basic = mandatory for procedure registration. Advanced = recommended but optional. N/A = not part of the procedure: its value can only be known once a session has run.
Structure — session record	How one measurement session reports the item.
Session-Level Tier	Read-Only = must follow the value given in the procedure. Editable = may vary within the range the procedure allows; the session records the value used. Basic = the value comes from the session itself and must be reported. Advanced = the value comes from the session itself, and reporting it is recommended.

Basic	Required
Advanced	Recommended
Read-Only	Fixed by the procedure
Editable	Adjustable within the bounds the procedure sets
N/A	Not applicable at that level

How to read the structure cells

Each structure cell is an instruction, written in one of these patterns:

Give one value for each X.	One value for every entry in a list. "For the procedure" or "for the session" means nothing repeats: give a single value. "One or more values" means several may be given, but nothing else repeats over them.
For each A, give one value for each B.	Two independent lists crossed: one value for every pairing, with A as the outer list. Analytical Precision reads: for each "Secondary Reference Material", give one value for each "Reported Variable".
For each A, give one value for each of its B entries.	Nesting: B exists only inside A. For each "Sample Name", give one value for each of its "Sampling Unit Name" entries — analysis points exist only inside a sample.
List the ...	The row is the list itself. Other rows are given for each of its entries, so it has to be filled in first.

Names in quotes refer to the lists below, each defined by one field in the TAPP. In the procedure column, a list that exists only in a session — samples, analysis points, combined results — drops out: the procedure gives one value, and each sample or point in a session inherits it.

Sample Name	the list of samples analysed in the session
Sampling Unit Name	the list of analysis points or areas, per sample
Target Material	the list in "Target Material" — the material types the procedure analyses
Target Species	the list of elements the procedure determines
Monitored Element	the list in "Monitored Elements" — what is measured to determine those elements
Reported Variable	the list in "Reported Variables and Units"
Secondary Reference Material	the list in "Secondary Reference Materials"
Combined Result	the list in "Combined Results" — each reported value that combines several individual results

Review criteria — one per comment column

Needed?	Is this item relevant to EPMA as you practise it? Is it one thing (otherwise Split), and distinct from other items (otherwise Merge)? Is it correctly named (otherwise Rename, and say how in Other comments)? Options: Keep · Drop · Merge with another row · Split into several rows · Rename.
Structure right? (procedure)	Does the procedure need one value, one value for each entry of a list (for example, per "Target Material"), or a list of its own? Options: Yes · Repeats over the wrong thing · Should be a list · Should be a single value.
Tier right? (procedure)	Can the value be known before any session? If not: Should not be in the procedure record. If so, is it essential for judging data quality or reproducing the work? Then: Should be required; otherwise: Should be optional. Options: Yes · Should be required · Should be optional · Should not be in the procedure record.
Structure right? (session)	Does a session report it once, or once per sample, per analysis point, per element…? Options: Yes · Repeats over the wrong thing · Should be a list · Should be a single value.
Tier right? (session)	Must the session copy the procedure's value unchanged (Should be fixed by the procedure)? May it vary within the procedure's bounds (Should be adjustable)? Or is it known only from the session (Should be required / Should be optional)? Options: Yes · Should be required · Should be optional · Should be fixed by the procedure · Should be adjustable.
Other comments	Anything else: the wording of Description or Example, missing items, wrong or missing modes, suggestions on the TAPP's structure.

Notes

"Session" is used throughout for what the TAPP itself calls the analysis level, because "analysis" is easily read as one analysis point, which is the one thing the record is not. "Session-Level Tier" here is the TAPP's Analysis-Level Tier.

This workbook is generated from EPMA_TAPP_v88.csv and is a snapshot of that version. Data types and allowed values are not shown; they are reviewed separately against the TAPP itself.

#	Metadata item	Needed?	Description	Example (NOT recommendations)	Structure — procedure record	Structure right?	Procedure-Level Tier	Tier right?	Structure — session record	Structure right?	Session-Level Tier	Tier right?	Other comments
1	**Group 1: Procedure Identification**												
1.1	**Procedure Name**	— Set by registration requirements; no review needed	A short descriptive name with a version number, identifying the instrument, the technique and the scope of what is measured.	EML-EPMA-CC v2, EPMA-WDS+EDS major and minor elements in carbonaceous-chondrite silicates, oxides, sulfides, phosphates and carbonates	Give one value for the procedure.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Read-Only	— Set by registration requirements; no review needed	
1.2	**Technique**	— Set by registration requirements; no review needed	Whether the procedure uses WDS, EDS or both.	EPMA-WDS+EDS (combines wavelength-dispersive and energy-dispersive spectrometry)	Give one value for the procedure.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Read-Only	— Set by registration requirements; no review needed	
1.3	**Procedure Author**	— Set by registration requirements; no review needed	The person or laboratory responsible for the procedure; ORCID recommended for a person.	Example Microanalysis Laboratory	Give one value for the procedure.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Read-Only	— Set by registration requirements; no review needed	
1.4	**Laboratory**	— Set by registration requirements; no review needed	The laboratory or institution hosting the instrument.	Example Microanalysis Laboratory	Give one value for the procedure.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Editable	— Set by registration requirements; no review needed	
1.5	**Laboratory ID**	— Set by registration requirements; no review needed	The laboratory's persistent identifier.	https://ror.org/0example00	Give one value for the procedure.	— Set by registration requirements; no review needed	Advanced	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Editable	— Set by registration requirements; no review needed	
1.6	**Procedure Start Date**	— Set by registration requirements; no review needed	The date this procedure configuration first went into production use.	January 2025	Give one value for the procedure.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Read-Only	— Set by registration requirements; no review needed	
1.7	**Funding Source for Procedure Development**	— Set by registration requirements; no review needed	The funding that supported the instrument, major upgrades and development of the procedure, with agencies and grant numbers.	Example Agency award EX-INST-0001 (instrument); award EX-DEV-0002 (procedure development)	Give one value for the procedure.	— Set by registration requirements; no review needed	Advanced	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Read-Only	— Set by registration requirements; no review needed	
1.8	**Procedure Reference(s)**	— Set by registration requirements; no review needed	Publications or technical reports describing, validating or benchmarking the procedure.	Example et al. (2025), doi:10.0000/example.2025.001	Give one value for the procedure.	— Set by registration requirements; no review needed	Advanced	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Read-Only	— Set by registration requirements; no review needed	
1.9	**Procedure DOI**	— Set by registration requirements; no review needed	The DOI of the registered procedure followed, or 'pending'.	doi:10.0000/example.epma-cc.v2	—	— Set by registration requirements; no review needed	N/A	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	
1.10	**Session Identifier**	— Set by registration requirements; no review needed	The laboratory's own run, sequence or batch identifier for the session, as generated by the instrument or acquisition software.	EML-2026-0412-A	—	— Set by registration requirements; no review needed	N/A	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	
1.11	**Analyst**	— Set by registration requirements; no review needed	The analyst or analysts who performed the session; ORCID recommended.	Analyst A (ORCID 0000-0000-0000-0000)	—	— Set by registration requirements; no review needed	N/A	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	
1.12	**Analysis Start Date**	— Set by registration requirements; no review needed	The date the session began.	2026-04-12	—	— Set by registration requirements; no review needed	N/A	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	
1.13	**Analysis End Date**	— Set by registration requirements; no review needed	The date the session ended.	2026-04-14	—	— Set by registration requirements; no review needed	N/A	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	
1.14	**Funding Source for Analysis**	— Set by registration requirements; no review needed	The funding that supported these analyses, with agencies and grant numbers.	Example Agency award EX-SCI-0003	—	— Set by registration requirements; no review needed	N/A	— Set by registration requirements; no review needed	Give one value for the session.	— Set by registration requirements; no review needed	Basic	— Set by registration requirements; no review needed	
1.15	**Coupled Technique(s)**		Other techniques applied to the same samples whose results are interpreted together with these, with the instrument, laboratory or purpose of each; 'None' if not coupled.	SEM; NanoSIMS (oxygen isotopes of carbonates, Example Isotope Laboratory)	Give one or more values for the procedure.		Advanced		Give one or more values for the session.		Editable		
1.16	**Coupling Description**		What data or context passes between the techniques, and which technique is performed first and why. Required whenever a coupled technique is named.	EPMA carbonate compositions select the matrix-matched reference material for the NanoSIMS mass-fractionation correction; SEM first, EPMA next (non-destructive), NanoSIMS last (sputtering destroys the analysed volume)	Give one value for the procedure.		Advanced		Give one value for the session.		Editable		
1.17	**Coupled Procedure DOI**		The registered procedure DOI of the coupled technique, a publication DOI, 'pending', or 'None'.	pending	—		N/A		Give one value for the session.		Advanced		
1.18	**Coupled Dataset or Publication Reference**		The identifier of the dataset or publication reporting both datasets together, 'same submission', or 'pending'.	same submission	—		N/A		Give one or more values for the session.		Advanced		
2	**Group 2: Samples**												
2.1	**Target Material**		The material types the procedure is designed to analyse.	Silicate mineral (olivine, pyroxene); oxide (magnetite, chromite, ilmenite); sulfide (pyrrhotite, pentlandite); phosphate (apatite, merrillite); carbonate (calcite, dolomite, breunnerite), in carbonaceous-chondrite material	List the material types the procedure is designed to analyse.		Basic		List the material types the procedure is designed to analyse.		Read-Only		
2.2	**Sample Preparation Method**		The form in which samples are presented to the instrument and the preparation that brought them to it; 'None' where material is analysed as received.	Epoxy mount, dry-polished with diamond to 0.25 µm, carbon coated to 20 nm, for EX-CC-01; the same, plus Ar-ion polishing after the final diamond step, for EX-CC-02	Give one value; each "Sample Name" entry in a session inherits it.		Basic		Give one value for each "Sample Name".		Editable		
2.3	**Sample Name**		The name of each sample analysed in the session, as the laboratory records it.	EX-CC-01; EX-CC-02	—		N/A		List every sample analysed in the session.		Basic		
2.4	**Sampling Unit Type**		The kind of subdivision of a sample that one row of reported results corresponds to, giving both levels where such units nest.	Analysis point (one point analysis = one result), labelled by particle and point number	Give one value for the procedure.		Basic		Give one value for the session.		Read-Only		
2.5	**Sampling Unit Name**		The label of each unit analysed and the sample it belongs to; where units are too numerous to name individually, the acquisition area they belong to.	P1-01–P1-12, P2-01–P2-11, P3-01–P3-06 for EX-CC-01 (29 points); P1-01–P1-16, P2-01–P2-08 for EX-CC-02 (24 points)	—		N/A		For each "Sample Name", list the analysis points analysed in it.		Basic		
2.6	**Target Material of Sampling Unit**		The Target Material entry each analysis point belongs to, which links it to the beam conditions registered for that material.	Silicate mineral for P1-01–P1-12, oxide for P2-01–P2-06, sulfide for P2-07–P2-11, phosphate for P3-01–P3-06 (EX-CC-01); carbonate for P1-01–P1-16, silicate mineral for P2-01–P2-08 (EX-CC-02)	—		N/A		For each "Sample Name", give one value for each of its "Sampling Unit Name" entries.		Basic		
2.7	**Sample Persistent Identifier**		The persistent identifier of the sample, at the level actually analysed; IGSN recommended.	IGSN:EXAMPLE0001 for EX-CC-01; IGSN:EXAMPLE0002 for EX-CC-02	—		N/A		Give one value for each "Sample Name".		Advanced		
2.8	**Sampling Unit Selection Criteria**		The criteria by which the units analysed are selected within a sample.	Grains at least 5 µm across (10 µm for phosphates and carbonates); at least 3 µm from boundaries, cracks and inclusions in BSE (5 µm for defocused or rastered beams); phase confirmed from its EDS spectrum first	Give one value for the procedure.		Basic		Give one value for the session.		Editable		
2.9	**Pre-Analysis Imaging and Screening**		Imaging or screening performed before analysis to locate and select the units analysed, with the technique, instrument and settings, and how individual analyses are linked back to the images.	BSE imaging and EDS maps (Mg, Si, Fe, Ca, S, P) on a field-emission SEM at 15 kV and 1 nA, registered to stage coordinates, each point labelled on its particle's BSE image — for both EX-CC-01 and EX-CC-02	Give one value; each "Sample Name" entry in a session inherits it.		Advanced		Give one value for each "Sample Name".		Editable		
3	**Group 3: Instrument & Software**												
3.1	**Instrument Manufacturer**		The manufacturer of the instrument.	JEOL	Give one value for the procedure.		Basic		Give one value for the session.		Read-Only		
3.2	**Instrument Model**		The model designation, including any generation or configuration suffix.	JXA-8530F Plus	Give one or more values for the procedure.		Basic		Give one or more values for the session.		Read-Only		
3.3	**Electron Source**		The type of electron source.	Schottky field-emission	Give one value for the procedure.		Basic		Give one value for the session.		Read-Only		
3.4	**Acquisition Software**		The software that controls the instrument and acquires the data, with version number.	Probe for EPMA v13 (Probe Software)	Give one value for the procedure.		Basic		Give one value for the session.		Editable		
3.5	**Data Processing Software(s)**		Every software package applied to the data after acquisition, with version numbers.	Probe for EPMA v13	Give one or more values for the procedure.		Basic		Give one or more values for the session.		Editable		

#	Metadata item	Needed?	Description	Example (NOT recommendations)	Structure — procedure record	Structure right?	Procedure-Level Tier	Tier right?	Structure — session record	Structure right?	Session-Level Tier	Tier right?	Other comments
3.6	**WDS Spectrometer Configuration**		The number, type and crystal range of the WDS spectrometers, with manufacturer and model.	Five JEOL spectrometers: LDE1/TAP (Sp1); TAP/PETJ (Sp2); PETJ/LIF (Sp3); PETH/LIFH (Sp4); LIF/PETL (Sp5)	Give one value for the procedure.		Advanced		Give one value for the session.		Read-Only		
4	**Group 4: Measurement Information**												
4.1	**Analytical Mode**		The analytical modes the procedure covers.	EDS point analysis; EDS mapping; WDS point analysis; WDS mapping	Give one or more values for the procedure.		Basic		Give one or more values for the session.		Read-Only		
4.2	**Beam Mode**		The beam mode for point analysis. Must be consistent with beam diameter and raster dimensions.	Focused for silicate minerals, oxides and sulfides; defocused for phosphates; rastered for carbonates	Give one value for each "Target Material".		Basic		Give one value for each "Target Material".		Read-Only		
4.3	**Accelerating Voltage**		The accelerating voltage in kV, with justification for any departure from the standard operating voltage.	15 kV (the laboratory's standard operating voltage)	Give one value for the procedure.		Basic		Give one value for the session.		Editable		
4.4	**Beam Current**		The probe current in nA for point analysis.	20 nA for silicate minerals, oxides and sulfides; 8 nA for phosphates; 4 nA for carbonates	Give one value for each "Target Material".		Basic		Give one value for each "Target Material".		Editable		
4.5	**Beam Diameter**		The beam diameter in µm for point analysis.	1 µm (focused) for silicate minerals, oxides and sulfides; 5 µm (defocused) for phosphates; focused within the raster for carbonates	Give one value for each "Target Material".		Basic		Give one value for each "Target Material".		Editable		
4.6	**Beam Raster Dimensions**		The width and height in µm of the area over which the beam is rastered.	5 × 5 µm for carbonates	Give one value for each "Target Material".		Advanced		Give one value for each "Target Material".		Editable		
4.7	**Beam Damage Minimization**		The measures taken to minimise beam damage, with the beam conditions used and the phases to which they are applied.	Defocused beam for phosphates; rastered beam for carbonates; Na, K, F and Cl measured in the first acquisition pass on every point	Give one value for each "Target Material".		Advanced		Give one value for each "Target Material".		Editable		
4.11	**Drift Correction**		How drift in beam current and spectrometer position is monitored and corrected during a session.	Beam current measured with the Faraday cup before each point and at the start of each map line, intensities normalised to it; primary standards re-measured at the start and end of the session, drift interpolated linearly with time	Give one value for the procedure.		Advanced		Give one value for the session.		Editable		
4.12	**Target Species**		The elements the procedure determines. Isotopes are never separate entries.	Si, Ti, Al, Cr, Fe, Mn, Mg, Ca, Na, K, P, S, F, Cl, Ni, Co, O, C (18 elements)	List the elements the procedure determines.		Basic		List the elements the procedure determines.		Editable		
4.13	**Monitored Elements**		The elements monitored in order to make the determinations, grouped under the determined element each serves; elements monitored only to correct an interference serve none.	Ti, Al, Cr, Mn, Na, K, P, F, Cl, Ni, Co; Zn (only to correct Zn Lα on Na Kα)	List each monitored element, grouped under the "Target Species" it serves (some serve none).		Basic		List each monitored element, grouped under the "Target Species" it serves (some serve none).		Read-Only		
4.14	**Reported Variables and Units**		The quantities the procedure finally reports, with their units, including intermediate quantities reported alongside final ones.	SiO_2, TiO_2, Al_2O_3, Cr_2O_3, FeO (total Fe), MnO, MgO, CaO, Na_2O, K_2O, P_2O_5, NiO, CoO in wt%; S, F, Cl in wt%; O equivalent of F and Cl; CO_2 by stoichiometry (carbonates); totals; structural formulas in apfu	List every quantity reported, with its unit.		Basic		List every quantity reported, with its unit.		Read-Only		
4.15	**X-ray Detection Method per Monitored Element**		Whether the monitored element is measured by WDS or by EDS.	WDS for Ti, Al, Cr, Mn, Na, K, P, F, Cl, Ni, Co and for Zn (interference monitor)	Give one value for each "Monitored Element".		Basic		Give one value for each "Monitored Element".		Read-Only		
4.16	**X-ray Line**		The X-ray emission line measured.	Kα for every WDS element, including Zn	Give one value for each "Monitored Element".		Basic		Give one value for each "Monitored Element".		Read-Only		
4.17	**Diffracting Crystal**		The diffracting crystal.	LDE1 for F; TAP for Na, Al; PETJ for K, P, Ti; PETH for Cl; LIF for Zn, Mn, Co; LIFH for Cr; LIFH + LIF for Ni	Give one value for each "Monitored Element".		Basic		Give one value for each "Monitored Element".		Read-Only		
4.18	**WDS Spectrometer Channel**		The spectrometer position or positions on which the element is measured, one entry per assignment, including an element measured on more than one spectrometer with the intensities aggregated.	Sp1 for F, Al; Sp2 for Na, P; Sp3 for K, Ti, Co; Sp4 for Cl, Cr; Sp5 for Zn, Mn; Sp4 + Sp5 for Ni (intensities aggregated)	Give one value for each "Monitored Element".		Advanced		Give one value for each "Monitored Element".		Read-Only		
4.19	**Sequence**		The order in which the element is acquired and, where the element suite exceeds the number of spectrometers, the acquisition pass to which it belongs.	Pass 1 for F, Na, K, Cl, Zn; pass 2 for Al, P, Ti, Cr, Mn; pass 3 for Co, Ni	Give one value for each "Monitored Element".		Advanced		Give one value for each "Monitored Element".		Read-Only		
4.20	**Proportional Counter / Detector**		The detector type.	Low-pressure P10 gas-flow counter for Sp1 and Sp2 (F, Al, Na, P); sealed Xe counter for Sp3–Sp5 (K, Ti, Co, Cl, Cr, Zn, Mn, Ni)	Give one value for each "Monitored Element".		Advanced		Give one value for each "Monitored Element".		Read-Only		
4.21	**WDS PHA Setting**		The pulse height analyser mode.	Differential for F, Na; integral for all other elements	Give one value for each "Monitored Element".		Advanced		Give one value for each "Monitored Element".		Read-Only		
4.22	**Peak Counting Time**		The on-peak counting time in seconds.	20 s for F, Cl, Al, P; 10 s for Na, K, Zn; 30 s for Ti, Cr, Mn; 60 s for Co, Ni (40 s in the registered procedure)	For each "Target Material", give one value for each "Monitored Element".		Basic		For each "Target Material", give one value for each "Monitored Element".		Editable		
4.23	**Background Counting Time**		The total off-peak background counting time in seconds, summed across all background positions.	20 s for F, Cl, Al, P; 10 s for Na, K, Zn; 30 s for Ti, Cr, Mn; 60 s for Co, Ni (total of both positions)	For each "Target Material", give one value for each "Monitored Element".		Basic		For each "Target Material", give one value for each "Monitored Element".		Editable		
4.24	**Background Position(s)**		The location of each background position relative to the peak, in millimetres or sin θ and on the high- or low-energy side, or that no off-peak positions are used.	−3.2 / +4.5 mm for F; −4.0 / +4.0 mm for Na, P; −4.5 / +4.5 mm for K; −3.0 / +3.0 mm for Cl, Cr, Ni; −3.5 / +3.5 mm for Zn, Al, Mn; −5.0 / +5.0 mm for Ti; −2.5 / +3.5 mm for Co	Give one value for each "Monitored Element".		Advanced		Give one value for each "Monitored Element".		Editable		
5	**Group 5: Data Processing**												
5.1	**Matrix Correction Method**		The matrix correction algorithm, which also applies when count maps are converted to concentration maps.	Armstrong/Love–Scott φ(ρz) matrix correction (Armstrong 1995) in Probe for EPMA	Give one value for the procedure.		Basic		Give one value for the session.		Read-Only		
5.2	**Mass Absorption Coefficients (MACs)**		The database of mass absorption coefficients the correction uses.	FFAST (Chantler et al. 2005)	Give one value for the procedure.		Basic		Give one value for the session.		Read-Only		
5.3	**X-ray Background Correction Method**		How the background beneath the peak is estimated and subtracted.	Two-point off-peak, interpolated linearly for all WDS elements except F; exponential fit for F	Give one value for each "Monitored Element".		Basic		Give one value for each "Monitored Element".		Editable		
5.4	**Time-Dependent Intensity Correction**		The time-dependent intensity correction applied.	Linear for Na; exponential for F (in apatite); none for other elements	Give one value for each "Target Species".		Advanced		Give one value for each "Target Species".		Editable		
5.5	**Target Species Estimation Method**		How the concentration of the element is obtained from the measured data.	Directly from measured intensities for every measured element; by stoichiometry from cation valences for O; as CO_2 by stoichiometry for C (carbonates only); OH in apatite by difference, F + Cl + OH = 1 apfu (Ketcham 2015)	Give one value for each "Target Species".		Advanced		Give one value for each "Target Species".		Read-Only		
5.6	**Halogen Correction on Oxygen**		Whether oxygen is adjusted for halogen substitution where oxygen is calculated by stoichiometry.	Oxygen equivalent of F and Cl subtracted from the total (apatite)	Give one value for the procedure.		Advanced		Give one value for the session.		Editable		
5.7	**WDS Dead Time Correction**		The dead-time correction algorithm and any instrument-specific constant. No measured WDS dead time is reported.	Logarithmic expression; spectrometer-specific constants of 1.1–1.5 µs	Give one value for the procedure.		Basic		Give one value for the session.		Read-Only		
5.9	**Blank Correction**		The method and reference material used to determine and subtract a blank.	Apparent concentration measured on Ni- and Co-free synthetic forsterite subtracted, for Ni and Co; none for other elements	Give one value for each "Target Species".		Advanced		Give one value for each "Target Species".		Editable		
5.10	**Normalization / Standards-Based Correction**		Any normalisation applied beyond the primary calibration, or 'None'.	Structural formulas normalised to a fixed anion basis: 4 O for olivine and spinels, 6 O for pyroxene and dolomite, 3 O for calcite, 13 anions for apatite (Ketcham 2015); none for other reported variables	Give one value for each "Reported Variable".		Advanced		Give one value for each "Reported Variable".		Editable		
5.11	**Calibration Factor and Determination Method**		Any externally calibrated factor that converts the measured quantity into the reported quantity, how that factor was determined, and its uncertainty.	None, for every reported variable (concentrations from k-ratios against the primary standards)	Give one value for each "Reported Variable".		Basic		Give one value for each "Reported Variable".		Editable		
5.12	**Procedural Blank Level**		The blank level measured in the session, together with the blank's composition where the reported quantity is a ratio.	35 ± 9 µg/g for Ni; 22 ± 8 µg/g for Co (n = 5)	—		N/A		Give one value for each "Target Species".		Basic		

#	Metadata item	Needed?	Description	Example (NOT recommendations)	Structure — procedure record	Structure right?	Procedure-Level Tier	Tier right?	Structure — session record	Structure right?	Session-Level Tier	Tier right?	Other comments
5.13	**Analysis Inclusion and Rejection Criteria**		The rules deciding which individual results contribute to a combined result, and the outcome of applying them: how many results were obtained, how many were included, and on what grounds any were excluded.	Rules: totals 98.5–101.5 wt% (silicates, oxides, sulfides), 96.0–101.5 (apatite), 98.0–102.0 (carbonates); cation sum within ±0.03 apfu; no overlap with a second phase; beam drift under 1%. Outcome: 53 points acquired, 45 included; excluded P3-02 and P3-05 from apatite in EX-CC-01 and P1-03, P1-07 from dolomite in EX-CC-02 (totals), P2-07 and P2-09 from pyrrhotite in EX-CC-01 and P1-16 from calcite in EX-CC-02 (overlap), P2-05 from olivine in EX-CC-02 (drift)	Give one value; each "Combined Result" entry in a session inherits it.		Basic		Give one value for each "Combined Result".		Basic		
5.14	**Combination Method**		Whether the reported variable is an individual result or combines several; if combined, the statistical model used and what the results are combined over.	Arithmetic mean of the included points, per phase within each sample, for every oxide and element concentration; maps and structural formulas not combined	Give one value for each "Reported Variable".		Basic		Give one value for each "Reported Variable".		Editable		
5.15	**Combined Results**		Each reported value obtained by averaging or otherwise combining several individual results, named with what it combines and the results that contribute to it.	Olivine in EX-CC-01 (P1-01–P1-08); pyroxene in EX-CC-01 (P1-09–P1-12); magnetite in EX-CC-01 (P2-01–P2-06); pyrrhotite in EX-CC-01 (P2-07–P2-11); apatite in EX-CC-01 (P3-01–P3-06); dolomite in EX-CC-02 (P1-01–P1-10); calcite in EX-CC-02 (P1-11–P1-16); olivine in EX-CC-02 (P2-01–P2-08)	—		N/A		List every reported value that combines several individual results, saying what each combines.		Basic		
5.16	**Constants and Reference Values Used**		The physical constants and citable reference values used in data reduction, with their sources, or 'None'.	IUPAC 2021 standard atomic weights (Prohaska et al. 2022)	Give one value for the procedure.		Basic		Give one value for the session.		Editable		
6	**Group 6: QC & Uncertainty**												
6.1	**Primary Calibration Standard Name**		The primary reference material against which the instrument is calibrated, its source, and a citation for the accepted values used; where quantification uses a stored library or response model, that instead. 'None' means no calibration was performed.	Synthetic SrF_2 for F; Amelia albite for Na; microcline USNM 143966 for K; scapolite USNM R6600-1 for Cl; anorthite USNM 137041 for Al; Wilberforce fluorapatite for P; ilmenite USNM 96189 for Ti; synthetic Cr_2O_3 for Cr; rhodonite for Mn; Co metal for Co; Ni metal for Ni; synthetic ZnS for Zn (Smithsonian values: Jarosewich et al. 1980)	For each "Target Material", give one value for each "Target Species".		Basic		For each "Target Material", give one value for each "Target Species".		Editable		
6.2	**Secondary Reference Materials**		The reference materials measured as unknowns alongside the samples, with the source of each and a citation for its accepted values.	San Carlos olivine USNM 111312/444; Kakanui augite USNM 122142; Durango apatite USNM 104021; chromite USNM 117075; calcite USNM 136321 (accepted values: Jarosewich et al. 1980)	List the secondary reference materials run as unknowns.		Basic		List the secondary reference materials run as unknowns.		Editable		
6.3	**X-ray Line Overlap Corrections Applied**		Whether an X-ray line overlap correction is applied and, if so, the overlapping lines and the correction method.	Yes for Co (Fe Kβ on Co Kα), Mn (Cr Kβ on Mn Kα) and Na (Zn Lα on Na Kα), quantitative and iterated with the matrix correction; no for all other elements	Give one value for each "Target Species".		Basic		Give one value for each "Target Species".		Read-Only		
6.4	**Interfering Elements**		The elements whose lines overlap the measured peak.	Fe for Co; Cr for Mn; Zn for Na	Give one value for each "Target Species".		Advanced		Give one value for each "Target Species".		Read-Only		
6.5	**Interference Correction Standard**		The reference material used to calibrate the interference correction.	Co-free Rockport fayalite for Co; Mn-free synthetic Cr_2O_3 for Mn; Na-free synthetic ZnS for Na	Give one value for each "Target Species".		Advanced		Give one value for each "Target Species".		Read-Only		
6.6	**Detection Limit**		The detection limit, with units, stating whether the value is a typical estimate or a session-specific measurement. Must be consistent with the stated method.	0.055 wt% for F; 0.018 for Na_2O; 0.009 for K_2O; 0.010 for Cl; 0.012 for Al_2O_3; 0.020 for P_2O_5; 0.015 for TiO_2; 0.018 for Cr_2O_3; 0.016 for MnO; 0.010 for CoO; 0.008 for NiO	Give one value for each "Reported Variable".		Advanced		Give one value for each "Reported Variable".		Basic		
6.7	**Detection Limit Method**		The formula or approach used to calculate the detection limit, with a citation where one exists.	3σ of the background counts, for every reported variable	Give one value for each "Reported Variable".		Basic		Give one value for each "Reported Variable".		Read-Only		
6.8	**Analytical Precision**		The reproducibility of repeated measurements, as a one-sigma relative standard deviation in percent, with the number of analyses and the measured value.	0.4–0.7% for oxides above 10 wt%; 0.9–1.6% for FeO and Al_2O_3; 2–4% for TiO_2, Na_2O, F and Cl; 4% for NiO in San Carlos olivine (1σ RSD; San Carlos olivine n = 9, Kakanui augite n = 6, Durango apatite n = 6, chromite n = 5, calcite n = 6)	For each "Secondary Reference Material", give one value for each "Reported Variable".		Advanced		For each "Secondary Reference Material", give one value for each "Reported Variable".		Basic		
6.9	**Analytical Accuracy**		The offset between measured and accepted values, as a percent relative bias, with the source of the accepted value.	Within ±1.2% relative for all major oxides; within ±2.6% for minor oxides and halogens	For each "Secondary Reference Material", give one value for each "Reported Variable".		Advanced		For each "Secondary Reference Material", give one value for each "Reported Variable".		Basic		
6.10	**Counting Statistics Error**		The uncertainty predicted from counting statistics, including the counts on any background or blank subtracted, with the sigma level stated.	1σ from counting statistics on the peak and on any subtracted background or blank, for every analysis and every reported variable; about 10% relative for NiO at 0.03 wt% in olivine	Give one value for each "Reported Variable".		Advanced		For each "Sample Name", give one value for every "Reported Variable" at every "Sampling Unit Name" entry.		Basic		
6.12	**Goodness-of-Fit or Dispersion Statistic**		The statistic used to assess whether scatter among the individual results contributing to a combined result exceeds what analytical uncertainty alone predicts, together with its value.	MSWD of the included points about the mean: for olivine in EX-CC-01, 0.9 for NiO to 2.3 for CaO, and 4.6 for FeO (Fe–Mg zoning); likewise for every other combined result	—		N/A		For each "Combined Result", give one value for each "Reported Variable".		Basic		
6.13	**Other Statistics**		Any further statistic reported for a combined result, with the statistic named and its value.	Standard error of the mean (1 s.d./√n): for olivine in EX-CC-01, 0.08 wt% for SiO_2, 0.12 for MgO, 0.04 for FeO; likewise for every other combined result	—		N/A		For each "Combined Result", give one value for each "Reported Variable".		Advanced		
6.14	**Additional Notes**		Any information not captured elsewhere, including anomalies, deviations from the registered procedure and instrument modifications.	Particle P2 of EX-CC-02 partly plucked during ion polishing; its analyses lie on the remaining fragment. No other anomalies, instrument modifications or departures	Give one value for the procedure.		Advanced		Give one value for the session.		Advanced		